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Department of Computer Science and Engineering
DevOps Lab - L3

Lab Assignment No: 9	
Name of Student	Aryan Sheladia
PRN	23070122202
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Class	CSE – B1
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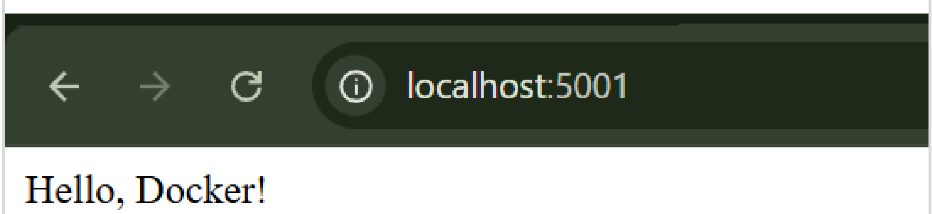
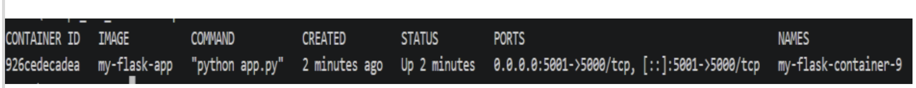
Title of Assignment	Container Management & Networking

Implementation & Theory

A Docker container was started from the existing my-flask-app image in detached mode using the -d option. Host port 5001 was mapped to container port 5000 using -p 5001:5000, allowing the Flask application to be accessed through the host machine. This follows the reference procedure, which uses port mapping 5001:5000.



The Flask application was accessed through the host machine using http://localhost:5001. The successful display of **"Hello, Docker!"** confirms that the container is running and the host-to-container port mapping is functioning correctly.



The docker ps command was used to display currently running containers. The output confirms that the Flask container is in the **Up** state and that port 5001 on the host is mapped to port 5000 inside the container.

```
PS E:\DevOps_SEM_7> docker stop my-flask-container-9
my-flask-container-9
PS E:\DevOps_SEM_7>
```

The running Docker container was stopped using docker stop. Docker returned the container name after successfully stopping it. Stopping a container terminates its running process while preserving the container so that it can still be inspected or removed later.

```
PS E:\DevOps_SEM_7> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS E:\DevOps_SEM_7> docker ps -a
CONTAINER ID   IMAGE                       COMMAND                  CREATED          STATUS                        PORTS     NAMES
926cedecadea   my-flask-app                "python app.py"          8 minutes ago    Exited (137) About a minute ago         my-flask-container-9
48d66661140e   my-flask-app                "python app.py"          24 minutes ago   Exited (137) 8 minutes ago              my-flask-container
cc111857475a   jenkins/jenkins:lts-jdk17   "/usr/bin/tini -- /u…"   9 days ago       Exited (143) 9 days ago                 myjenkins
9f64d7378874   qdrant/qdrant               "./entrypoint.sh"        2 weeks ago      Exited (143) 2 days ago                 qdrant
d88b7ca7114e   neo4j                       "tini -g -- /startup…"   2 weeks ago      Exited (137) 2 days ago                 neo4j
PS E:\DevOps_SEM_7>
```

docker ps displays only currently running containers, while docker ps -a displays all containers, including stopped ones. After stopping the Flask container, it no longer appeared in docker ps, but remained in docker ps -a with an **Exited** status.

```
E:\DevOps_SEM_7> docker inspect my-flask-container-9
        },
        "NetworkSettings": {
            "Bridge": "",
            "SandboxID": "",
            "SandboxKey": "",
            "Ports": {},
            "HairpinMode": false,
            "LinkLocalIPv6Address": "",
            "LinkLocalIPv6PrefixLen": 0,
            "SecondaryIPAddresses": null,
            "SecondaryIPv6Addresses": null,
            "EndpointID": "",
            "Gateway": "",
            "GlobalIPv6Address": "",
            "GlobalIPv6PrefixLen": 0,
            "IPAddress": "",
            "IPPrefixLen": 0,
            "IPv6Gateway": "",
            "MacAddress": "",
            "Networks": {
                "bridge": {
                    "IPAMConfig": null,
```

The docker inspect command was used to examine the configuration of the stopped container. The NetworkSettings section provides information such as the container's internal IP address, gateway, and

	configured port bindings. This helps verify how the container is connected to the Docker network.
	
	`PS E:\DevOps_SEM_7> docker rm my-flask-container-9`
	`my-flask-container-9`
	`PS E:\DevOps_SEM_7> docker ps -a`
	`CONTAINER ID   IMAGE                       COMMAND                  CREATED          STATUS                        PORTS     NAMES`
	`48d66661140e   my-flask-app                "python app.py"          27 minutes ago   Exited (137) 10 minutes ago             my-flask-container`
	`cc111857475a   jenkins/jenkins:lts-jdk17   "/usr/bin/tini -- /u…"   9 days ago       Exited (143) 9 days ago                 myjenkins`
	`9f64d7378874   qdrant/qdrant               "./entrypoint.sh"        2 weeks ago      Exited (143) 2 days ago                 qdrant`
	`d88b7ca7114e   neo4j                       "tini -g -- /startup…"   2 weeks ago      Exited (137) 2 days ago                 neo4j`
	`PS E:\DevOps_SEM_7>`
	
	The stopped container was permanently removed using docker rm. The docker ps -a command was then used to verify that the container no longer exists. This demonstrates the difference between stopping a container, which preserves it, and removing a container, which deletes it.